

Question

The following chemical processes exist:

1. Cobalt chloride desiccant changes from blue to pink in air.
2. HMTA reacts with nitric acid to produce RDX (cyclotrimethylenetrinitramine).
3. Hydrogen peroxide is added to a $\text{pH} = 0$ TiCl_4 solution, and the solution turns red. The titanium-containing cation in the product has a titanium mass fraction of 54.5%.
4. Acetone reacts with sulfuryl chloride, producing only one small molecule.
5. Potassium permanganate is added dropwise to a ferrous chloride solution, and a fleeting blue color is observed. The product Mn^{2+} is not to be written in the equation for this process.

Regarding the sum of the coefficients of all substances in the chemical/ionic equations of the above reactions (ionic equations for processes 3 and 5, chemical equations for processes 1, 2, and 4, coefficients are balanced to the simplest integer ratio), there are a total of five numbers. Which of the following statements about these five numbers is correct:

- A. All other options are incorrect
- B. All five numbers are less than 10.
- C. All five numbers are greater than 5.
- D. Among the five numbers, there are no prime numbers.
- E. Among five numbers, there exist two integer power exponents of 2.
- F. Among five numbers, there exist two multiples of 5.

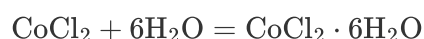
G. Among five numbers, there is no multiple of 3.

Answer

Correct Answer: **E**

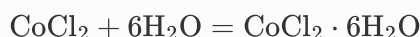
Detailed Explanation

Process 1: It is obvious that this is the water absorption reaction of anhydrous cobalt chloride. According to chemical common sense, the complex is hexahydrate. The equation is as follows:



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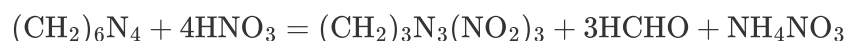
The sum of coefficients is 8.

Process 2: The chemical formula of RDX is $(\text{CH}_2)_3\text{N}_3(\text{NO}_2)_3$. Obviously, this reaction is a nitration reaction. Considering the preparation of urotropine, it is known that the excess carbon atoms are converted into formaldehyde, and the nitrogen atoms are converted into ammonium ions. Therefore, the equation can be balanced:

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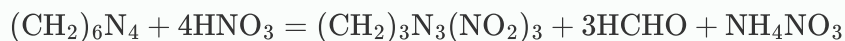
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Considering the preparation of urotropine, it is known that the excess carbon atoms are converted into formaldehyde, and the nitrogen atoms are converted into ammonium ions



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The sum of coefficients is 10.

Process 3: According to the knowledge of elemental chemistry, the red color should be the color of the peroxo complex of Ti(IV); considering the existence form, combined with the mass fraction, it is known that $\text{Ti}_2\text{O}_5^{2+}$ is generated. Balance the ion equation:

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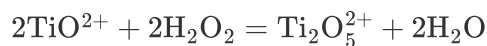
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The red color should be the color of the peroxo complex of Ti(IV)

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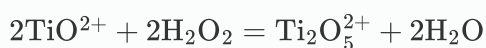
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Combined with the mass fraction, it is known that $\text{Ti}_2\text{O}_5^{2+}$ is generated



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The sum of coefficients is 7.

Process 4: Referring to the reaction in organic chemistry, sulfuryl chloride can convert the ketone carbonyl group into a dichloro substituent. Since only one small molecule is produced, α -halogenation is excluded. Therefore, the equation can be written as:

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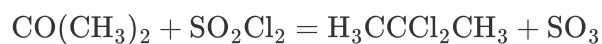
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Sulfuryl chloride can convert the ketone carbonyl group into a dichloro substituent

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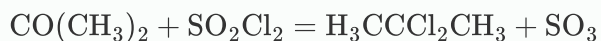
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Only one small molecule is produced, α -halogenation is excluded



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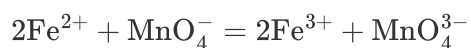
The sum of coefficients is 4.

Process 5: According to the knowledge of elemental chemistry, the blue color is the intermediate product of oxidation, MnO_4^{3-} ; thus, the equation is balanced:

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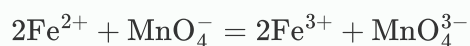
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The blue color is the intermediate product of oxidation, MnO_4^{3-}



CHECKPOINT

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The sum of coefficients is 6.

Therefore, the five numbers are 8, 10, 7, 4, and 6, indicating that only option E is correct.

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The five numbers are 8, 10, 7, 4, and 6